

VALUE INTELLIGENCE PLATFORM

Build Plan & Execution Document

A digital decision assistant for SME entrepreneurs. Detailed design, system architecture, tech stack, and a phased execution plan built around realistic token budgets per session.

"What is my company worth today, what drives that value, and which actions could grow it over time?"

VERSION	v1.0 — Design & Architecture
AUDIENCE	Team · Faculty · Future engineers
COURSE	Master in Data Science for Management — Cattolica
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DOCUMENT MAP

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01 · EXECUTIVE SUMMARY

A decision assistant, not a calculator.

VIP is a digital decision assistant for SME entrepreneurs. It estimates current enterprise value, scores strategic quality across four capitals, identifies the value drivers and destroyers in the business, and recommends the three highest-impact actions to grow value over the next 24–36 months.

The product is built around a single insight: most existing tools either describe the past (accounting and BI) or apply a naive multiple to EBITDA. Neither helps an entrepreneur decide what to *do*. VIP closes that gap by combining a small set of carefully-chosen quantitative inputs with a short qualitative questionnaire, and translating the result into a transparent valuation and a prioritised action plan.

What this document is

A complete design and execution plan. It captures the product concept, the four-capital model, the valuation formula and its calibration, the questionnaire structure, the five platform engines, the database schema, the algorithmic pipeline, the visual design system, and a ten-phase roadmap to build the platform — each phase scoped to fit within one coding session.

Success criteria

Clarity of problem	A reader unfamiliar with the project understands the entrepreneur's need within 60 seconds.
Model robustness	The four-capital scoring is defensible per-input; weights and ranges are documented and justified.
Originality	The model goes beyond EBITDA × multiple — incorporating Value Gap, ROV-ranked actions, and a simulation engine.
Feasibility	A working web prototype exists; the data pipeline runs end-to-end on the AIDA calibration dataset.
Output quality	The dashboard collapses the result into one page the entrepreneur can act on.
Completeness	Pitch deck, prototype, executive summary, build plan — all delivered.

02 · PRODUCT CONCEPT

The strategic intelligence gap.

SMEs do not lack data — ERP, CRM, accounting systems already produce more than most entrepreneurs realise. What they lack is a translation layer: from data into a quantified, navigable picture of company value and the levers that drive it.

This translation layer is the space VIP occupies. It is not management software, not BI reporting, and not a financial-advisory service. It is a self-serve digital product that produces strategic insight from a small, deliberately-chosen set of inputs.

The three gaps

Data is everywhere	Storage and capture are no longer the bottleneck. Every SME has at least one ERP, one accounting package, and increasingly one CRM.
Insight is missing	Nobody translates the data into a clear picture of value and the levers that drive it. Reporting describes; it does not decide.
Decisions in the dark	Investment, exit, and growth decisions are committed without a quantified compass. Entrepreneurs rely on instinct or a single advisor's view.

Five capabilities the platform must deliver

#	Capability	What it means in practice
01	Estimate value	A credible methodology calibrated to real SMEs — not a naive EBITDA × multiple.
02	Score quality	Multi-dimensional scoring across the four capitals, normalised to peer group.
03	Surface drivers	Which factors create value. Which silently erode it. Transparently named.
04	Recommend actions	Top 3 highest-impact levers over 24–36 months, ranked by Return on Value.
05	Deliver simply	One dashboard, designed for the entrepreneur — not the data scientist.

03 · THE 4-CAPITAL MODEL

An SME's value is the sum of four capitals.

The model decomposes strategic quality into four distinct, observable lenses. Each is scored independently and then aggregated into a single Strategic Quality Factor (SQF) that feeds the valuation formula.

Weights are tuned for manufacturing — Technological Capital is intentionally over-weighted versus a services framework. Weights can be re-tuned for other sectors.

Capital	Weight	What it captures	Key drivers
Financial	35%	How the business performs financially today: profitability, growth, liquidity, leverage, cash quality.	EBITDA · Revenue growth · Recurring revenue · Free cash flow · Client concentration
Technological	20%	The hard infrastructure that underwrites scalability: plant, machinery, IP, digital maturity.	Digital maturity · Automation · Proprietary data · Enabling systems · IP
Human & Organisational	25%	Workforce productivity, governance depth, founder dependency, transferability.	Management structure · Founder dependency · Processes · Culture · Transferability
Relational	20%	Network, partnerships, brand, ecosystem. Drives growth, market reach, and negotiating power.	Network quality · Partnerships · Brand · Ecosystem · Reputation

Why peer-relative percentiles

Manufacturing financials are wildly heterogeneous. A specialty parts maker and a large engineered-systems manufacturer can both legitimately sit in the same NACE code with very different absolute numbers. Percentile-rank within the peer group is outlier-robust, sub-sector aware, and interpretable: "top quartile on profitability, bottom quartile on liquidity" is a sentence anyone can act on.

04 · VALUATION LOGIC

One transparent formula. No black boxes.

Core formula

$$V = \text{EBITDA} \times M_{\text{sector}} \times \text{SQF} \times \text{GF}$$

EBITDA	Normalised 3-year average operating performance, smoothed for one-off items.
M_{sector}	Sector multiple, drawn from Damodaran or peer transactions, discounted for SME size and illiquidity.
SQF — 0.6 to 1.4	Strategic Quality Factor. Aggregated four-capital score. Penalises fragility, rewards scalability.
GF — 0.7 to 1.5	Growth Factor. Expected CAGR plus quality of growth (organic vs episodic), normalised to sector and size.

Value Gap — the platform's differentiating insight

Value Gap = Optimised Potential Value – Current Value.

The distance the entrepreneur could close by executing the recommended actions. This is what distinguishes a calculation tool from a decision assistant: not just *what is your company worth today*, but *how much value is sitting on the table* and which levers unlock it.

05 · INPUT ENGINE

The questions the model knows how to ask.

Around twenty carefully-chosen inputs, organised in three families. The smartest model is not the one with the most variables — it is the one that selects the most relevant ones. Every input below either feeds a capital score or anchors the valuation to a peer group.

Qualitative · 1–5 questionnaire

Question	What it asks	Maps to
Founder dependency	How much does the business rely on you personally?	Human & Org
Management structure	Depth and quality of the leadership team	Human & Org
Digital maturity	Automation, proprietary data, enabling systems	Technological
Client portfolio quality	Diversification, contract length, stickiness	Relational
Business model scalability	Can revenue grow without proportional cost?	Technological / Financial
Network & partnerships	Strategic relationships, ecosystem position	Relational

Quantitative · hard numbers

Input	Why we collect it
Revenue · last 3 years	For CAGR computation and 3-year EBITDA averaging
EBITDA & margins	Core operating performance signal
% recurring revenue	Quality of revenue — feeds Financial & Relational
Top-3 client concentration (%)	Fragility flag — feeds Relational and SQF penalty
Revenue CAGR	Direct input to Growth Factor
Tech investment / revenue ratio	Proxy for Technological Capital investment intensity

Contextual · where you sit

Input	Why we collect it
Sector / vertical	Determines the peer group and sector multiple base
Lifecycle stage	Adjusts Growth Factor expectations
Distinctive assets	Modulates Technological & Relational scores
M&A / exit history	Anchors Value Gap interpretation
Stated objective & horizon	Time-weights the recommended actions

06 · FIVE ENGINES

Platform anatomy.

The platform is composed of five named, inspectable modules. Each produces an artefact that the next consumes. Nothing is hidden inside a single end-to-end black box.

#	Engine	What it does	Output
01	Input Engine	Captures quantitative submissions and the qualitative questionnaire (~20 inputs).	JSON payload per company per submission
02	Scoring Model	Computes per-capital scores, overall quality, risk index, scalability index — all peer-relative.	4 capital scores + composite + indices
03	Valuation Logic	Combines EBITDA, sector multiple, SQF and GF into V. Computes Value Gap.	Current V, Potential V, Value Gap
04	Recommendation Engine	Surfaces drivers and destroyers; ranks improvement actions by Return on Value (ROV).	Top 3 prioritised actions with V uplift
05	Simulation Engine	Interactive what-if: how does V move if I lift this lever by 20%? Updates dashboard in real time.	Side-by-side V comparison · upgrade module

The Simulation Engine is the conceptual upgrade

The base brief asks for Input, Scoring, Valuation, and Recommendation. We add a fifth: an interactive simulation layer where the entrepreneur can move sliders on individual levers (e.g. client concentration, recurring revenue %) and watch the valuation update live. This converts the platform from a one-time diagnostic into an ongoing planning tool.

07 · DASHBOARD & UX

The decision, on one page.

The entrepreneur opens VIP at 11pm with a teaser in their inbox and 20 minutes before they need to call back. The dashboard exists to make that 20 minutes count. Every metric the platform produces collapses into one screen.

What the dashboard shows

Element	Example value	Why it's on the page
Estimated Value	€4.2M (range €3.8M – €4.7M)	Headline — what the company is worth today
Value Gap	+38% optimised potential	How much value is on the table
Quality Score (CQS)	67 / 100	Above-average structure, peer-relative
Risk Index	MEDIUM · client concentration flag	Highest-priority fragility
4-Capital Radar	Fin 68 · Tech 54 · H&O 71 · Rel 55	Where strengths and gaps live
Top 3 Actions	+12% · +9% · +7% V uplift	What to do next, ranked by ROV

Design principles

- **One screen, no scroll.** If the entrepreneur has to scroll to see Value Gap, the design has failed.
- **Three numbers > thirty.** Estimated Value, Value Gap, Risk Index — everything else is supporting evidence.
- **Diagnosis → prescription.** Every flagged risk has a recommended action attached.
- **Defensible by default.** Every score is reachable to its inputs in two clicks; nothing is mysterious.
- **Cinematic but functional.** Motion serves comprehension. No animation for animation's sake.

08 · TECH STACK

Modern, hosted, free to start, scale-ready.

The stack is chosen so a two-to-four-person team can ship a working prototype in phases, with zero infrastructure cost in development. Everything has a generous free tier and a clear paid path when the demo lands.

Layer	Choice	Why
Frontend framework	Next.js 14 (App Router)	Server components for data fetching, edge runtime where useful, file-based routing, hot reload.
Language	TypeScript	Strict mode; types generated from the Supabase schema via <code>supabase gen types</code> .
Styling	Tailwind CSS + CSS variables	Mirrors the design tokens from the cinematic site; works with our utility-first speed.
Motion	Framer Motion (UI) + GSAP & ScrollTrigger (scroll cinematics) + Lenis (smooth scroll)	Framer for component-level animation. GSAP/ScrollTrigger for pinning & scrubbed reveals. Lenis for the buttery scroll feel.
Charts	Recharts + custom SVG for the radar	Recharts for stock series and bar charts. The 4-capital radar stays custom for visual control.
Icons	Lucide React	Consistent stroke, tree-shakeable.
Database	Supabase · Postgres	Postgres with managed extensions. Row-level security on every user-scoped table.
Auth	Supabase Auth	Email-OTP, OAuth (Google), JWT in cookies. RLS policies tie data to <code>auth.uid()</code> .
API	Next.js Route Handlers + Supabase Edge Functions	Route Handlers for simple reads; Edge Functions for the scoring pipeline.
Data ingestion	Python · pandas · openpyxl · psycpg / supabase-py	Reuses the existing <code>split_aida_capitals.py</code> outputs; loads into the five tables.
Hosting	Vercel (frontend) + Supabase Cloud (backend)	Free dev tier. Production: Pro plans, ~€30/mo combined.
CI / CD	GitHub Actions	Lint, type-check, build on every PR. Auto-deploy from main.
Observability	Vercel Analytics + Supabase Logs + Sentry (Phase 9)	Performance, error tracking, and database query insight.

09 · SYSTEM ARCHITECTURE

Data flow, top to bottom.

Component layers

Presentation	Next.js App Router · React server & client components · Tailwind · Framer Motion · GSAP · Lenis · Recharts. Hosted on Vercel's edge network.
Application	Route Handlers in /app/api/* for thin REST endpoints. Supabase Edge Functions for the scoring pipeline (Deno runtime).
Data	Supabase Postgres. Five capital tables for the AIDA calibration set + user-facing tables for submissions, valuations, and recommendations. RLS scopes every read.
Auth	Supabase Auth · email-OTP + Google OAuth. JWT in HTTP-only cookies. SSR-aware via @supabase/ssr.
Ingestion	Python script consumes the AIDA Excel split and bulk-inserts into the five calibration tables. Runs once during setup; idempotent.
Observability	Vercel Analytics for web vitals. Sentry for client errors. Supabase logs for SQL performance.

Request flow — dashboard load

1. User authenticates via Supabase Auth (email-OTP or Google).
2. Next.js server component fetches the user's company_profile + latest valuation row.
3. If the valuation is stale (or missing), the server route calls the scoring Edge Function.
4. Edge Function runs the six-stage pipeline against the user's submission, persists the result.
5. Server component renders the dashboard with the four-capital radar, KPIs, and Top-3 actions.
6. Simulation Engine (client component) hydrates with the baseline and listens to slider changes.

Why Edge Functions for scoring

The pipeline is small (~36 metrics, peer-relative ranks, weighted sums) but it touches several tables. Putting it close to Postgres keeps latency low and lets us reuse it for the Simulation Engine without round-trips through the client.

10 · DATABASE SCHEMA

Five calibration tables, five user tables.

Calibration set · loaded once from AIDA

Table	What it stores	Notes
context	Identifiers, sector codes, peer group, size estimate	Joined to every per-capital row by company tax code
financial_capital	Profitability, growth, liquidity, leverage, cash quality columns	40 fields per company
technological_capital	Fixed assets, R&D, IP, capex intensity columns	40 fields per company
human_organisational	Workforce, productivity, governance, group structure columns	33 fields per company
relational_capital	Customers, suppliers, brand, internationalisation columns	23 fields per company

User-facing set · created per company per user

Table	What it stores	Notes
profiles	User profile, plan, organisation	Linked to auth.users via 1:1
companies	A user's own company (or a target they're evaluating)	RLS: only the owner reads/writes
submissions	A snapshot of the entrepreneur's questionnaire + quantitative inputs	Immutable; new submission = new row
valuations	Computed: V, V_potential, Value Gap, Quality Score, Risk Index, capital scores	Result of the scoring pipeline
recommendations	Top-3 actions, with ROV scores, capital deltas, and time horizon	Linked to a valuation row

RLS policies — the short version

Every user-facing table carries a `user_id` column. The default policy is `USING (user_id = auth.uid())` for `SELECT`, `UPDATE`, `DELETE`. Inserts are gated by `WITH CHECK (user_id = auth.uid())`. Service-role inserts (e.g. the scoring function) bypass RLS but are gated by signed function calls.

11 · SCORING PIPELINE

Six stages. Each one inspectable.

The pipeline produces a named, inspectable artefact at every stage. If a number on the dashboard looks wrong, we can trace it back to the input that drove it.

#	Stage	What it does	When / where
01	Ingestion	Consolidate AIDA exports on tax code → 14,999 × 153 calibration dataset.	Once · Python ETL
02	Capital split	Partition the wide table into five typed files; context columns appear in every file.	Once · Python (existing)
03	Metric engineering	~36 ratios, CAGRs, stability indicators derived per company (e.g. Debt/EBITDA, Recurring Revenue %, Top-3 client concentration).	Per submission · Edge Function
04	Peer benchmarking	Convert each metric to a 0–100 percentile within the company's AIDA peer group.	Per submission · Edge Function
05	Aggregation	Within-capital weighted sum → 4 capital scores → composite CQS (Capital Quality Score).	Per submission · Edge Function
06	Valuation	Map CQS → SQF, compute GF from CAGR + growth quality, look up M_{sector} , multiply through.	Per submission · Edge Function

Worked example — ACME INDUSTRIE S.R.L.

$EBITDA_{\text{norm}}$	€4.2M (3-yr avg, smoothed for one-offs)
M_{sector}	5.8× (NACE 282, illiquidity-discounted)
SQF	1.15 (above-average four-capital score)
GF	1.20 (moderate growth, organic)
V	$4.2 \times 5.8 \times 1.15 \times 1.20 = \text{€}33.6\text{M}$
$V_{\text{potential}}$	€46.4M if Top-3 actions executed
Value Gap	+38% / +€12.8M

12 · ROV & SIMULATION ENGINE

From diagnosis to prescription, then to live planning.

Return on Value (ROV) — the prioritisation logic

ROV ranks candidate actions by their expected impact on enterprise value over a 24–36 month horizon, normalised by effort and feasibility. For each candidate action the engine computes:

ΔSQF or ΔGF	How much the action moves the Strategic Quality or Growth Factor.
Estimated ΔV	The resulting enterprise-value delta = $V \times ((SQF' / SQF) \times (GF' / GF) - 1)$.
Effort score	A 1–5 score capturing how reachable the action is (team capacity, capital required).
Time-to-impact	Months until the action shows up in the financials.
ROV	$ROV = \Delta V / \text{Effort}$, time-weighted. Top-3 by ROV become the recommendations.

Example actions and impacts

#	Action	Target shift	Capital impact	ΔV
1	Reduce client concentration	60% top-3 → <40%	SQF +0.12	+12% V
2	Introduce recurring revenue	Multi-year / subscription	GF +0.15	+9% V
3	Strengthen middle management	Reduce founder dependency	Transferability low → high	+7% V

Simulation Engine — the upgrade

Sliders for every lever the entrepreneur controls. Move "Top-3 client concentration" from 60% to 40% and the dashboard re-renders in real time: new Value Gap, new SQF, new ROV ranking. The simulation is bounded by the same formulas as the baseline, so the projections are always defensible.

13 · VISUAL DESIGN SYSTEM

Premium, calm, intelligent.

Tone & references

The aesthetic is luxury consulting meets futuristic fintech. Awwwards-tier cinematic storytelling but without the gaming-style neon overload. Inspiration: Stripe, Linear, Vercel, Mattermark, Brex marketing. Reference dark mode dominant on marketing surfaces; product UI lighter for sustained reading.

Colour system

Token	Hex	Role
Ink	#0F172A	Primary text on light, primary surface on dark
Gold	#F5A524	Headline accent · KPIs · brand mark
Cyan	#06B6D4	Eyebrow labels · architecture flow
Purple	#A855F7	Questionnaire accent · Technological Capital
Green	#22C55E	Value Gap · positive deltas · Relational Capital
Red	#EF4444	Risk Index · destroyer flags
Blue	#3B82F6	Financial Capital
Orange	#F97316	Human & Organisational Capital

Typography

Display / Serif accents	Fraunces — italic for headline accent, contemporary serif for emotion
UI Sans	Inter — body and UI, with stylistic sets enabled (ss01)
Monospace	JetBrains Mono — numbers, code, eyebrows, formula

Motion principles

Purposeful	Animation serves comprehension. Numbers count up because counting up makes them feel measured.
Cinematic	Sequenced scroll reveals — never all at once.

Calm

Easing is mostly power3.out. No bouncing UI.

Performant

GSAP timelines on the marketing surface. Framer Motion variants in the product.
Reduced-motion respected.

14 · PHASED EXECUTION PLAN

Ten phases. Each fits one session.

Each phase below is scoped to be deliverable in a single coding session (~30–60k output tokens of code + verification). Phases declare their inputs, deliverables, acceptance criteria, and a rough token estimate. They are intentionally small enough that one failed session does not blow up the project plan.

PHASE 00

▪ Foundation

DONE

Repo scaffolding, AIDA splitter, calibration data, design infographic.

Inputs	AIDA exports · project brief
Deliverables	• Git repository with data/, docs/, src/, README.md • split_aida_capitals.py producing 5 typed Excel files • Static + cinematic infographic websites • SME_Valuation_Design.pdf (the original design doc)
Acceptance	Repo pushable to GitHub; all .xlsx files generated; cinematic site opens locally.
Token estimate	– (already complete)

PHASE 01

▪ Next.js project skeleton

NEXT

Bootstrap the production codebase. Establish design tokens, layout, fonts.

Inputs	Cinematic single-file site as visual reference.
Deliverables	• Next.js 14 App Router project with TypeScript, Tailwind, ESLint • globals.css with CSS variables matching the design system • Layout shell: top brand, side-nav dots, scroll progress • Marketing route group with placeholder pages • README updated with run instructions
Acceptance	npm run dev serves the skeleton at localhost:3000 with the dark theme applied.
Token estimate	~40k

PHASE 02 ▪ Marketing surface port

Port the cinematic single-file site to React components, scene by scene.

Inputs	Phase 1 skeleton; index.html as visual reference.
Deliverables	• , , , , , , , , , , • GSAP + ScrollTrigger + Lenis integration via a useEffect hook • Particle canvas extracted into a client component • Responsive breakpoints preserved
Acceptance	Marketing site at / matches the single-file version visually and behaviourally.
Token estimate	~60k

PHASE 03 ▪ Supabase setup & ingestion

Stand up the database. Load the 14,999-SME calibration set.

Inputs	AIDA Excel files in data/.
Deliverables	• Supabase project + local CLI configured • Migration SQL for the 5 calibration tables + 5 user-facing tables • RLS policies for user-facing tables • Python ingestion script that loads the calibration set • TypeScript types generated via supabase gen types
Acceptance	Querying <code>SELECT count(*) FROM financial_capital</code> returns 14,999.
Token estimate	~50k

PHASE 04 ▪ Auth & onboarding

Sign-up, login, first-company creation, persisted session.

Inputs	Phase 3 schema.
Deliverables	• Supabase Auth integrated via @supabase/ssr • /login, /signup, /verify pages • /onboarding wizard that creates a company row • Auth-aware layout (server component reads session) • Middleware that redirects unauthenticated users from /dashboard
Acceptance	A new user can sign up, verify by email-OTP, create a company, and land on an empty dashboard.
Token estimate	~50k

PHASE 05 · Questionnaire UI

Multi-step form that collects the 20 inputs and writes a submission.

Inputs	Phase 4 auth; companies table.
Deliverables	• Step 1: Quantitative numbers (6 fields, currency/percentage formatting) • Step 2: Qualitative questionnaire (6 1–5 questions with animated dot scale) • Step 3: Contextual inputs (5 fields) • Step 4: Review + submit • Form state via React Hook Form + Zod schema • INSERT into submissions on submit
Acceptance	A submission row exists in Supabase with all 17 fields populated.
Token estimate	~55k

PHASE 06 · Scoring & valuation Edge Function

The six-stage pipeline as a Supabase Edge Function.

Inputs	Phase 3 schema; Phase 5 submissions.
Deliverables	• Edge Function score-submission (Deno) • Metric engineering — ~36 derived metrics • Peer benchmarking — percentile rank within sector • Within-capital aggregation → 4 scores → SQF • Growth Factor + sector multiple lookup → V • Value Gap computation against optimised potential • Persistence to valuations table
Acceptance	Calling the function with a submission_id returns and persists a complete valuation row.
Token estimate	~60k

PHASE 07 · Dashboard shell & KPIs

The /dashboard page — KPI grid, company header, 4-Capital radar (static), Top-3 actions placeholder.

Inputs	Phase 6 valuations.
Deliverables	• /dashboard/[companyId] route • with Estimated Value, Value Gap, Quality Score, Risk Index • custom-SVG component, animated polygon morph • with status pill • Skeleton states + empty states
Acceptance	Dashboard renders ACME-style values from the user's own valuation row.
Token estimate	~55k

PHASE 08 · Recommendation engine & ROV

Generate and persist Top-3 actions; render them on the dashboard.

Inputs	Phase 6 pipeline outputs.
Deliverables	• Action catalogue (seed): client concentration, recurring revenue, mgmt depth, etc. • ROV scoring function (ΔV / Effort, time-weighted) • Persistence to recommendations • dashboard panel with ROV-ranked list
Acceptance	Top-3 actions appear on the dashboard with V uplift values and the action description.
Token estimate	~45k

PHASE 09 · Simulation Engine + polish + deploy

Interactive what-if sliders, performance pass, Vercel deploy.

Inputs	Phase 6 + 7 + 8.
Deliverables	• with sliders per lever • Client-side recomputation that mirrors the Edge Function logic for speed • Side-by-side "current vs simulated" V • Lighthouse pass on /dashboard ≥ 90 across the four pillars • Sentry wiring • Vercel preview + production
Acceptance	Public URL exists; moving any slider updates the V value in <100ms.
Token estimate	~55k

15 · TOKEN-BUDGET STRATEGY

How we stay inside one session per phase.

Every phase is scoped to fit inside a single Claude coding session. The strategy is a combination of (a) keeping each phase narrow, (b) carrying forward only the minimal context needed for the next phase, and (c) making each phase self-verifying so we do not waste tokens debugging across boundaries.

Per-session budget

Phase brief	~1–2k · the relevant section of this document
Existing-code context	~5–10k · only the files the phase touches
New code output	~25–45k · the actual code produced this session
Verification	~5k · bash sanity checks, type checks, smoke tests
Headroom	~5–10k · for the inevitable follow-up question
Total per session	~50–70k tokens

Carry-forward artefacts

Between phases, only three things need to survive: the database schema (as SQL), the TypeScript types (auto-generated), and the design tokens (CSS variables). Everything else can be rebuilt from this document.

Rule of thumb

If a phase requires >60k tokens of new code output, split it. Phase 02 (marketing port) is the largest in this plan and may be split into 02a (Hero + Problem + Capabilities + Capitals) and 02b (Formula + Input + Engines + Dashboard + Architecture + Closing) if needed.

16 · RISKS & MITIGATIONS

What could go wrong, and how we cope.

Risk	Mitigation	Likelihood
Data quality — missing AIDA columns	The splitter is whitespace-tolerant and prints a per-capital missing-column report; missing fields default to peer-group median during scoring.	LOW
Auth complexity / OAuth edge cases	Use Supabase Auth out-of-box; email-OTP first, OAuth in a later phase. No custom auth.	LOW
Simulation realism	Start with linear deltas calibrated to historical AIDA peer movements. Iterate towards non-linear feedback once a baseline exists.	MED
Performance at 14,999 rows	Peer-percentile lookups are pre-computed nightly and cached per peer group. Edge Functions query the cached materialised view.	LOW
SQF / GF calibration drift	Ranges are anchored to the AIDA distribution; recalibrate quarterly using a held-out validation set.	MED
Demo-day failure mode	Pre-seed three example companies (one strong, one fragile, one mid). Demo never depends on live network > LAN.	LOW
Scope creep ("can we add a fifth capital?")	The four-capital structure is in the brief and the design doc. Treat additions as v2.	LOW

17 · DELIVERABLES CHECKLIST

Mapped to the project-work brief.

Deliverable	Where it lives	Status
A · Problem definition	The entrepreneur's need and the gap from existing tools — sections 01 & 02 of this document; opening of the cinematic site.	DONE
B · Logical model design	Value dimensions, variables, indicators — section 03; 4-Capital section of the site.	DONE
C · Questionnaire structure	Questions, collection method, mapping to model variables — section 05; Input Engine section of the site.	DONE
D · Algorithmic logic	Input → Score → Value estimate; weight construction and justification — section 11.	DONE
E · Platform output	What the user sees; which indicators; how results are visualised — section 07; Dashboard section of the site.	DONE
F · Improvement levers	How priority actions are identified and their value impact estimated — section 12.	DONE
G · Product design	Dashboard mockup, UX, how it stays simple yet powerful — section 07; cinematic site mockup.	DONE
Pitch presentation (25–30 min)	TBD — will reuse sections + dashboard demo.	PENDING
Concept deck (max 10 slides)	TBD — distillation of this document.	PENDING
Prototype / interactive mockup	index.html (cinematic site) covers the mockup requirement; live prototype lands in Phase 09.	IN PROGRESS
Executive summary (1 page)	TBD — one-page condensation for an investor or entrepreneur.	PENDING

18 · APPENDICES

Reference material for the team.

A · Glossary

AIDA	Italian financials database by Bureau van Dijk — our calibration set.
CQS	Capital Quality Score — composite four-capital score, 0–100.
EBITDA	Earnings before interest, taxes, depreciation, amortisation.
GF	Growth Factor — 0.7–1.5, scales V by growth trajectory and quality.
NACE	EU industry classification. NACE 28xx = Machinery & Equipment manufacturing.
Peer group	AIDA-assigned grouping by sub-sector + size band (e.g. 282 VL).
ROV	Return on Value — ranking score for candidate actions.
SQF	Strategic Quality Factor — 0.6–1.4, scales V by four-capital quality.
Value Gap	Optimised Potential V minus Current V; the upside on the table.
V	Enterprise value, computed as $\text{EBITDA} \times M_{\text{sector}} \times \text{SQF} \times \text{GF}$.

B · Reference materials in the repo

docs/SME_Valuation_Design.pdf	Original design document (G. Tedeschi, 2026).
docs/VIP_Build_Plan.pdf	This document.
infographic.html	Static infographic (v2).
infographic-v1.html	Backup of static infographic.
index.html	Cinematic immersive website.
src/split_aida_capitals.py	Splitter script — the existing pipeline stage 02.
data/all_capitals_clean_split.xlsx	Combined 5-sheet workbook for the 14,999 SMEs.

C · Reading order for a new teammate

- README.md — repository overview & setup.
- docs/SME_Valuation_Design.pdf — the original model rationale.
- docs/VIP_Build_Plan.pdf — this document; design + execution plan.
- index.html — the cinematic site (visual reference for the product).
- src/split_aida_capitals.py — the existing pipeline stage.
- Phase 01 onwards — pick up the next phase from section 14.

"Don't design an algorithm. Design a decision assistant."

End of document · VIP · Build Plan v1.0